

1. Solucionar la ecuación diferencial

$$\left(\frac{dy}{dx}\right)^3 - 2y \left(\frac{dy}{dx}\right)^2 - xy \left(\frac{dy}{dx}\right)^2 + 2xy^2 \left(\frac{dy}{dx}\right) + 12x^2y^3 - 6x^2y^2 \left(\frac{dy}{dx}\right) = 0$$

Sea $z = \frac{dy}{dx}$

$$z^3 - 2y z^2 - xy z^2 + 2xy^2 z + 12x^2y^3 - 6x^2y^2 z = 0$$



$$z^3 + (-2y - xy) z^2 + (2xy^2 - 6x^2y^2) z + 12x^2y^3 = 0$$

Probando si $z = 2y$ es raíz

$$P(2y) = (2y)^3 + (-2y - xy)(2y)^2 + (2xy^2 - 6x^2y^2)(2y) + 12x^2y^3 = 0$$

$$8y^3 + (-2y - xy)(4y^2) + (2xy^2 - 6x^2y^2)(2y) + 12x^2y^3 = 0$$

$$8y^3 - 8y^3 - 4xy^3 + 4xy^3 - 12x^2y^3 + 12x^2y^3 = 0$$

$$0 = 0$$

Dividiendo $z = 2y$

2y	1	$-2y - xy$ $2y$	$2xy^2 - 6x^2y^2$ $-xy \cdot 2y$	$12x^2y^3$ 0
1	-xy	$-6x^2y^2$	0	

$$P(z) = (z - 2y)(z^2 - xy z - 6x^2y^2)$$

$$P(z) = (z - 2y)(z - 3xy)(z + 2xy) = 0$$

Resolviendo cada factor

$$z - 2y = 0 \quad \vee \quad z - 3xy = 0 \quad \sim \quad z + 2xy = 0$$

$\frac{dy}{dx} - 2y = 0$	$\frac{dy}{dx} - 3xy = 0$	$\frac{dy}{dx} + 2xy = 0$
$\frac{dy}{dx} = 2y$	$\frac{dy}{dx} = 3xy$	$\frac{dy}{dx} = -2xy$
$\int \frac{1}{y} dy = \int 2 dx$	$\int \frac{1}{y} dy = \int 3x dx$	$\int \frac{1}{y} dy = \int -2x dx$
$\ln y = 2x + C_1$	$\ln y = \frac{3}{2}x^2 + C_2$	$\ln y = -x^2 + C_3$

2. Determinar una solución continua del siguiente problema de valor inicial

$$(1+x^2) \frac{dy}{dx} + 2xy = f(x), \quad y(0) = 0$$

donde

$$f(x) = \begin{cases} x & \text{si } 0 \leq x < 1 \\ 1 & \text{si } x \geq 1 \end{cases}$$

$$(1+x^2) \frac{dy}{dx} + 2xy = f(x)$$

Reescribiendo la ecuación, dividiendo todo por $(1+x^2)$

$$\frac{dy}{dx} + \underbrace{\frac{2x}{1+x^2}}_{p(x)} y = \underbrace{\frac{f(x)}{(1+x^2)}}_{q(x)}$$

$$u(x) = e^{\int \frac{2x}{1+x^2} dx (*)}$$

$$* \int \frac{2x}{1+x^2} dx \quad \text{Sea } w = 1+x^2 \Rightarrow dw = 2x dx$$

$$\int \frac{1}{w} dw = \ln|w| + C = \ln|1+x^2| + C$$

$$u(x) = e^{\ln|1+x^2|} = 1+x^2$$

1) Para $0 \leq x < 1$

$$\frac{dy}{dx} + \frac{2x}{1+x^2} y = \frac{x}{1+x^2}$$

$$\frac{dy}{dx} + \frac{2x}{1+x^2} y = \frac{x}{1+x^2}$$

$$(1+x^2) y = \int \left(\frac{x}{1+x^2} \right) (1+x^2) dx + C$$

$$(1+x^2) y = \int x dx + C$$

$$(1+x^2) y = \frac{x^2}{2} + C.$$

$$y = \frac{1}{2(1+x^2)} (x^2 + C.)$$

2) Para $x \geq 1$

$$\frac{dy}{dx} + \frac{2x}{1+x^2} y = \frac{1}{1+x^2}$$

$$(1+x^2) y = \int \left(\frac{1}{1+x^2} \right) (1+x^2) dx + C$$

$$(1+x^2) y = \int 1 dx + C$$

$$(1+x^2) y = x + C$$

$$y = \frac{1}{1+x^2} (x + C_2)$$

de $y(0) = 0$

$$0 = \frac{1}{2(1+(0)^2)} (0^2 + C_1)$$

$$0 = \frac{C_1}{2} \Rightarrow C_1 = 0$$

Para la continuidad

$$\lim_{x \rightarrow x_0} f(x) = f(x_0)$$

$$\lim_{x \rightarrow 1^-} f(x) = f(1)$$

↳ Solo por la izquierda porque por la derecha ya lo subimos

$$\lim_{x \rightarrow 1^-} \frac{1}{1+x^2} (x+C_2) = \frac{x}{2(1+x^2)}$$

$$\frac{1}{2} (1+C_2) = \frac{1}{2+1}$$

$$1+C_2 = \frac{1}{3} (2)$$

$$C_2 = \frac{2}{3} - 1 = -\frac{1}{3} \Rightarrow$$

$$x - \frac{1}{3} \quad \frac{3x-1}{3(1+x^2)}$$

Solución continua

$$\begin{cases} \frac{x}{2+2x^2} & \text{si } 0 \leq x < 1 \\ \frac{x - \frac{1}{3}}{1+x^2} & \text{si } x \geq 1 \end{cases}$$

3. Encontrar la familia ortogonal a la familia $y + x \ln y = x \ln x + cx$

$$y + x \ln y - x \ln x - cx = 0$$

$$x \ln x = \ln x + \frac{1}{x} \cdot x$$

$$\frac{y + x \ln y - x \ln x}{x} = c$$

$$\ln \frac{y}{x}$$

$$\text{Cuando } \theta = 90^\circ \quad f' g' = -1$$

$$1 + \frac{x}{y} \quad \frac{y+x}{y}$$

$$f' = \frac{dy}{dx} = -\frac{F_x}{F_y} = -\frac{\ln y - \ln x + 1}{1 + \frac{x}{y}} = -\left(\frac{y + x \ln y - x \ln x}{x} \right)$$

multiplico por $\frac{x}{x}$

Nota

$$\frac{-x \ln y}{x} = \frac{x \ln x + x - y}{x} = \frac{x \ln x + x \ln x}{x}$$

$$\frac{-\cancel{x \ln y} = \cancel{x \ln x} + x - y = \cancel{x \ln y} + \cancel{x \ln x}}{x \left(1 + \frac{x}{y}\right)}$$

$$f' = - \frac{x - y}{x \left(1 + \frac{x}{y}\right)}$$

$$g' = \frac{dy}{dx} = -1 - \frac{1}{f'} = -1 \cdot \frac{x \left(1 + \frac{x}{y}\right)}{(x + y)}$$

$$\frac{dy}{dx} = - \frac{x \left(1 + \frac{x}{y}\right)}{(x + y)}$$

$$\frac{dy}{dx} = - \frac{\frac{xy + x^2}{y}}{\frac{x+y}{x+y}} = - \frac{\cancel{x(y+x)}}{\cancel{y(x+y)}} = - \frac{x}{y} = \frac{dy}{dx} \Rightarrow -x dx = y dy$$

$$\int -x dx = \int y dy$$

$$-\frac{x^2}{2} = \frac{y^2}{2} + C$$

$$-\frac{x^2}{2} - \frac{y^2}{2} = C$$

4. Solucionar la ecuación diferencial $\underbrace{\left(3xy + \frac{x^{-1}}{x^2y^2+1}\right)}_M dx + \underbrace{\left(2x^2 + \frac{y^{-1}}{x^2y^2+1}\right)}_N dy = 0$

Verificando si es exacta

$$\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} = 0$$

$$\frac{\partial M}{\partial y} = 3x + x^{-1} \cdot \frac{-2yx^2}{(x^2y^2+1)^2} = 3x - \frac{2yx}{(x^2y^2+1)^2}$$

$$\frac{\partial N}{\partial x} = 4x + y^{-1} \cdot \frac{-2xy^2}{(x^2y^2+1)^2} = 4x - \frac{2xy}{(x^2y^2+1)^2}$$

$$\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} = 3x - \frac{2yx}{(x^2y^2+1)^2} - 4x + \frac{2xy}{(x^2y^2+1)^2}$$

$$= -x$$

$$\bullet \frac{M_y - N_x}{N} = \frac{-x}{3xy + \frac{x^{-1}}{x^2y^2+1}}$$

$$\bullet \frac{M_y - N_x}{-M} = \frac{-x}{2x^2 + \frac{y^{-1}}{x^2y^2+1}}$$

$$\bullet M_y - N_x = nN x^{-1} - mM y^{-1}$$

$$-x = n \left(2x^2 + \frac{y^{-1}}{x^2y^2+1} \right) x^{-1} - m \left(3xy + \frac{x^{-1}}{x^2y^2+1} \right) y^{-1}$$

$$-x = n \left(2x + \frac{1}{yx(x^2y^2+1)} \right) - m \left(3x + \frac{1}{yx(x^2y^2+1)} \right)$$

$$-x = 2xn + \frac{n}{yx(x^2y^2+1)} - 3xm - \frac{m}{yx(x^2y^2+1)}$$

Iguando términos se tiene:

$$-1 = 2n - 3m \quad (1)$$

$$\frac{n}{yx(x^2y^2+1)} - \frac{m}{yx(x^2y^2+1)} = 0$$

$$\frac{n - m}{yx(x^2y^2+1)} = 0$$

$$n - m = 0 \quad (2)$$

Resolviendo el sistema por determinantes

$$n - \begin{vmatrix} -1 & -3 \\ 0 & -1 \end{vmatrix} \quad 1 = 0 \quad 1 \quad 1$$

RESOLUCION DE SISTEMA POR DETERMINANTES

$$n = \frac{\begin{vmatrix} -1 & -3 \\ 0 & -1 \end{vmatrix}}{\begin{vmatrix} 2 & -3 \\ 1 & -1 \end{vmatrix}} = \frac{1-0}{-2-(-3)} = \frac{1}{1} = 1$$

$$m = \frac{\begin{vmatrix} 2 & -1 \\ 1 & 0 \end{vmatrix}}{1} = \frac{0-(-1)}{1} = 1$$

$$N(x,y) = x^n y^m = xy$$

$$\left(3xy + \frac{x^{-1}}{x^2y^2+1}\right) dx + \left(2x^2 + \frac{y^{-1}}{x^2y^2+1}\right) dy$$

$$xy \left(3xy + \frac{x^{-1}}{x^2y^2+1}\right) dx + xy \left(2x^2 + \frac{y^{-1}}{x^2y^2+1}\right) dy = 0$$

$$\underbrace{\left(3x^2y^2 + \frac{y}{x^2y^2+1}\right)}_M dx + \underbrace{\left(2x^3y + \frac{x}{x^2y^2+1}\right)}_N dy = 0$$

$$\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} = 0$$

$$6x^2y + \frac{(1)(x^2y^2+1) - (y)(2yx^2)}{(x^2y^2+1)^2} = \frac{\partial M}{\partial y}$$

$$6x^2y + \frac{x^2y^2+1 - 2y^2x^2}{(x^2y^2+1)^2} = \frac{\partial M}{\partial y}$$

$$6x^2y + \frac{1 - x^2y^2}{(x^2y^2+1)^2} = \frac{\partial M}{\partial y}$$

$$6x^2y + \frac{(1)(x^2y^2+1) - (x)(2xy^2)}{(x^2y^2+1)^2} = \frac{\partial N}{\partial x}$$

$$6x^2y + \frac{x^2y^2+1 - 2x^2y^2}{(x^2y^2+1)^2} = \frac{\partial N}{\partial x}$$

$$6x^2y + \frac{1 - x^2y^2}{(x^2y^2+1)^2} = \frac{\partial N}{\partial x}$$

$$\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} = 0 \Rightarrow \cancel{6x^2y + \frac{1 - x^2y^2}{(x^2y^2+1)^2}} - \cancel{6x^2y + \frac{1 - x^2y^2}{(x^2y^2+1)^2}} = 0$$

$$\left(2x^3y + \frac{x}{x^2y^2+1}\right) dy$$

$$\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} = 0 \Rightarrow \cancel{6x^2y} + \frac{1 - x^2y^2}{(x^2y^2 + 1)^2} - \cancel{6x^2y} - \frac{1 + x^2y^2}{(x^2y^2 + 1)^2}$$

Por lo tanto es exacta

$$\exists F \text{ s.t. } \frac{\partial F}{\partial x} = M \quad \wedge \quad \frac{\partial F}{\partial y} = N$$

$$\underbrace{\left(3x^2y^2 + \frac{y}{x^2y^2 + 1}\right)}_M dx + \underbrace{\left(2x^3y + \frac{x}{x^2y^2 + 1}\right)}_N dy = 0$$

$$\int \frac{\partial F}{\partial y} = \int 2x^3y + \frac{x}{x^2y^2 + 1} + g(x)$$

$$u = x^2y^2 + 1$$

$$F(x, y) = x^3y^2 + \int \frac{x}{x^2y^2 + 1} + g(x)$$

Para $by^2 \pm a$ se sustituye $y = \frac{\sqrt{a}}{\sqrt{b}} u$

$$a=1; b=x^2 \Rightarrow y = \frac{1}{x} u \Rightarrow u = xy$$

$$dy = \frac{1}{x} du$$

$$x \int \frac{1}{x^2 \left(\frac{1}{x} u\right)^2 + 1} \cdot \frac{1}{x} du$$

$$x \int \frac{1}{\cancel{x^2} \left(\frac{u^2}{\cancel{x^2}}\right) + 1} \cdot \frac{1}{x} du$$

$$\frac{x}{x} \int \frac{1}{u^2 + 1} du$$

$$\arctan(u)$$

$$F(x, y) = x^3y^2 + \arctan(xy) + g(x)$$

$$\frac{\partial F}{\partial x} = M$$

$$3x^2y^2 + \frac{y}{1+x^2y^2} + g'(x) = 3x^2y^2 + \frac{y}{x^2y^2+1}$$

$$\int g'(x) dx = 0$$

$$g(x) = C$$

$$F(x,y) = x^3y^2 + \tan^{-1}(xy) + C \rightarrow \text{solución general}$$

7. Reduciendo el orden, solucionar la ecuación diferencial $x \frac{d^3y}{dx^3} - 12x^3 = 2 \frac{d^2y}{dx^2}$ con condiciones iniciales $y(1) = 1$, $y'(1) = 1$, $y''(1) = 0$

$$\text{Sea } z = \frac{d^2y}{dx^2} \Rightarrow \frac{dz}{dx} = \frac{d^3y}{dx^3}$$

$$x \frac{dz}{dx} - 12x^3 = 2z$$

$$x \frac{dz}{dx} - 2z = 12x^3 \rightarrow \text{E.D. primer orden en } z$$

divido todo por x teniendo en cuenta que solo es válido $x \neq 0$

$$\frac{dz}{dx} - \underbrace{\frac{2}{x} z}_{P(x)} = \underbrace{12x^2}_{Q(x)}$$

$$N(x) = e^{\int -\frac{2}{x} dx} = e^{-2 \ln x} = e^{\ln x^{-2}} = x^{-2}$$

$$x^{-2} \frac{dz}{dx} - 2x^{-3} z = 12$$

Se ve que la regla del producto inversa es $\frac{d}{dx}(u \cdot v) = u \frac{dv}{dx} + v \frac{du}{dx}$

Se ve que la regla del producto inversa es $\frac{d}{dx}(U \cdot V) = U \frac{dv}{dx} + v \frac{du}{dx}$
en este caso $U = z$ y $V = x^{-2}$

entonces

$$\frac{d}{dx}(z x^{-2}) = z \cdot \frac{d}{dx}(x^{-2}) + x^{-2} \cdot \frac{d}{dx}(z)$$

$$\frac{d}{dx}(z \cdot x^{-2}) = z \cdot -2x^{-3} + x^{-2} \frac{dz}{dx}$$

Por lo tanto

$$\int \frac{d}{dx}(z x^{-2}) dx = \int 12 dx$$

$$z x^{-2} = 12x + C_1$$

$$z = 12x^3 + C_1 x^2$$

Sabemos que $z = \frac{d^2 y}{dx^2}$ y por condición inicial $y''(1) = 0$

entonces

$$0 = 12(1)^3 + C_1(1)^2$$

$$0 = 12 + C_1 \Rightarrow C_1 = -12$$

entonces

$$\frac{d^2 y}{dx^2} = 12x^3 - 12x^2$$

entonces

$$\frac{dy}{dx} = \int 12x^3 - 12x^2 dx$$

$$\frac{dy}{dx} = 3x^4 - 4x^3 + C_2$$

Con $y'(1) = 1$ entonces

...

Con $y'(1) = 1$ entonces

$$1 = 3(1)^4 - 4(1)^3 + C_2$$

$$1 = 3 - 4 + C_2$$

$$1 = -1 + C_2$$

$$1 + 1 = C_2 \Rightarrow C_2 = 2$$

$$\frac{dy}{dx} = 3x^4 - 4x^3 + 2$$

$$y = \int 3x^4 - 4x^3 + 2 \, dx$$

$$y = \frac{3}{5}x^5 - x^4 + 2x + C_3$$

Con $y(1) = 1$

$$1 = \frac{3}{5}(1)^5 - (1)^4 + 2(1) + C_3$$

$$1 = \frac{3}{5} - 1 + 2 + C_3$$

$$1 = \frac{8}{5} + C_3 \Rightarrow C_3 = -\frac{3}{5}$$

entonces

$$y = \frac{3}{5}x^5 - x^4 + 2x - \frac{3}{5} \quad \rightarrow \text{solución}$$